

AP CALCULUS AB - UNIT 1

Guided Notes 1.12: Confirming Continuity over an Interval



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we distinguish continuity on open, closed, and half-open intervals?

- Distinguish continuity on open, closed, and half-open intervals.
- Use one-sided continuity at included endpoints.
- Find maximal intervals of continuity from domain restrictions.
- Justify continuity of sums, products, quotients, roots, and compositions on stated intervals.

Key Knowledge, Vocabulary, and Notation

Interior points	Use ordinary two-sided continuity at every interior point.
Included left endpoint	Require $\lim_{x \rightarrow a^+} f(x) = f(a)$.
Included right endpoint	Require $\lim_{x \rightarrow b^-} f(x) = f(b)$.
Maximal interval	An interval cannot be enlarged without crossing a domain restriction or discontinuity.
Domain first	Continuity of a standard function is asserted only on its domain.

Concept Development and Strategy

1. Find the real domain before writing intervals.
2. Split at denominator zeros, logarithm boundaries, even-root restrictions, and piecewise joins.
3. Use square brackets only when the endpoint is in the domain and the correct one-sided continuity condition holds.

Worked Example: Rational intervals

Find the maximal intervals of continuity of $f(x) = \frac{x+1}{x^2-4}$.

Answer and Reasoning

The denominator vanishes at ± 2 . The maximal intervals are $(-\infty, -2)$, $(-2, 2)$, and $(2, \infty)$.

Worked Example: Square-root endpoint

On what interval is $g(x) = \sqrt{5-x}$ continuous?

Answer and Reasoning

The domain is $x \leq 5$. Square root and the inner linear function are continuous, so g is continuous on $(-\infty, 5]$, with left continuity at 5.

Worked Example: Combined restrictions

Find maximal intervals for $h(x) = \frac{\ln(x+4)}{x^2-9}$.

Answer and Reasoning

Require $x > -4$ and $x \neq \pm 3$. Thus the intervals are $(-4, -3)$, $(-3, 3)$, and $(3, \infty)$.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Find the intervals of continuity of $f(x) = 1/(x - 5)$.

2. Find the interval of continuity of $g(x) = \sqrt{x + 2}$.

3. Find maximal intervals for $\frac{x^2+1}{(x-1)(x+4)}$.

4. Find the continuity set of $\sqrt{9 - x^2}$.

5. Find maximal intervals for $f(x) = \frac{\sqrt{x+1}}{x-3}$.

6. Find maximal intervals for $F(x) = \ln\left(\frac{x-2}{x+1}\right)$.

7. Prove that $\max\{f, g\}$ is continuous wherever f and g are continuous.

Multiple-Choice Check

1. The maximal intervals of continuity of $1/(x^2 - 9)$ are

- 2.** $\sqrt{4-x}$ is continuous on

- 3.** A calculator domain plot for $\sqrt{(x-1)/(x+2)}$ suggests real values on

- ## Common Errors and AP Precision

- ## Exit Ticket

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